

Barsha KC Khatri

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Education

Institute of Engineering, Pashchimanchal Campus	2021 – Present
• Bachelor's in Electronics, Communication and Information Engineering • Relevant Coursework: Data Structures Algorithms, Discrete Structure, Database Management, Artificial Intelligence, Data Mining, Big Data, Computer Networks, Operating Systems	

Experience

Data Science Intern , Tangible Careers	July 2024 - Oct 2024
• Developed Python-based regression models to forecast ride demand, applying real-world datasets. • Automated preprocessing pipelines using NumPy and Pandas to clean and transform data. • Collaborated using GitHub and Jupyter notebooks to document and present outcomes.	
Data Science Fellow , Code for Nepal	April 2024 - Feb 2025
• Gained experience in data cleaning, analysis, and visualization using Python (Pandas, Matplotlib, Seaborn). • Worked on sample projects such as fare prediction, customer segmentation, and sentiment analysis. • Improved understanding of real-world data workflows and presentation of results.	

Projects

Nepali Sign Language Recognition System

- Collaborated with a team to develop a real-time Nepali Sign Language recognition system using YOLOv8.
- Contributed to data preprocessing, model training, and testing for accurate gesture recognition.
- Tools Used: OpenCV, Python

Obstacle Detection System for Visually Impaired

- Collaborated with a team to develop a real-time obstacle detection system using YOLOv3 to assist visually impaired individuals in navigating their environment.
- Collected and annotated custom image datasets, fine-tuned YOLO model parameters to improve detection accuracy.
- Utilized OpenCV for real-time video feed processing and implemented YOLO integration for object detection.
- Conducted field testing under varying lighting conditions to ensure reliability of the system in real-world scenarios.

Annapurna Travel

- Collaborated with a teammate to develop and maintain back-end components using Django and Python for a travel website promoting tourism in the Annapurna region.
- Used Git and GitHub for version control, managing branches and resolving merge conflicts during collaborative development.
- Performed debugging and cross-browser testing to ensure functionality and user experience before deployment.

Technologies

Languages: C++, C, Python, HTML/CSS, SQL

Frameworks: Django

Libraries: pandas, NumPy, Matplotlib, seaborn, OpenCV

Achievements

BE Project Demonstration, MechTriX, Pulchowk Campus Winner

- Collaborated with a team of four to develop and showcase a robotic arm powered by machine learning, designed to efficiently load and manage smaller weights.

Project Demonstration, Octate 2024, Pashchimanchal Campus Winner

- Showcased our team project "Obstacle Detection System for the Visually Impaired" at an event named "Octate" organized in Pashchimanchal Campus.

Project Demonstration, Gandaki IT Mahotsav, 2081 2nd runner up

- Showcased our team project "Obstacle Detection System for the Visually Impaired" at Gandaki IT Mahotsav organized by CSITian Association of Pokhara, Nepal.

Certifications

Machine Learning Specialization

Coursera (DeepLearning.AI, Andrew Ng)

Issued: August 2023

Extracurricular

FemTech Club

June 2024 - Feb 2025

Former Vice-President, Pashchimanchal Campus

- Organized and hosted events including Ideathon, Graphic Design Workshop, and Design Hackathon.
- Managed event logistics, sponsorship outreach, and campus-wide promotions.
- Mentored junior members and contributed to fostering a collaborative tech community.

Robotics Club

2022 - 2024

Member, Pashchimanchal Campus

- Participated in technical discussions and hands-on sessions related to robotics and automation.
- Mentored junior members, guiding them in basic electronics, and Arduino-based projects.

C Programming Workshop

Mentor, Pashchimanchal Campus

- Guided beginners through fundamental concepts of C programming during a multi-day technical workshop organized by Terai Creative Committee of Pashchimanchal Campus.
- Contributed to curriculum design and hands-on session delivery.